

Predicting College Student Performance Using Machine Learning Techniques



Nathan Castro, Fatmir Gjergji

Faculty Advisor: Md Rafiul Islam, Ph.D.

University of the Incarnate Word

The 10th Coastal Bend Mathematics & Statistics Conference
Texas A&M University–Corpus Christi
April 18, 2026

Open University Learning Analytics Dataset

About Open University

- One of the largest universities in Europe with enrollment of 200,000
- Headquarters based in Milton Keynes, Buckinghamshire, England
- Popular for its flexible distance learning programs



About Open University Learning Analytics Dataset

- Contains student demographic, registration, assessment and daily interactions with courses in OU's Virtual Learning Environment from the years 2013 and 2014
- The data is anonymized
- This dataset contains 22 courses with 32,593 students

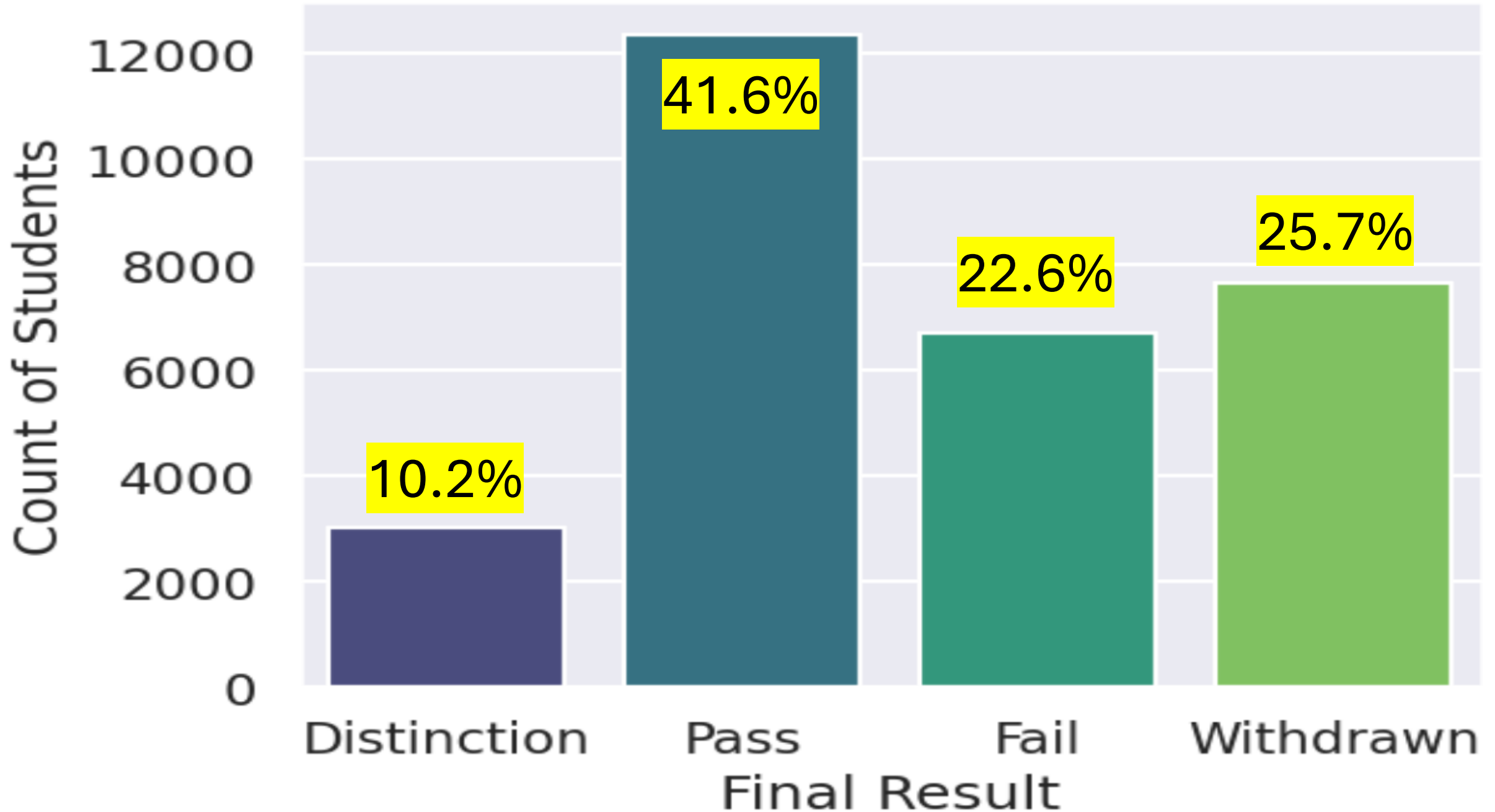
Exploratory Data Analysis

Objectives

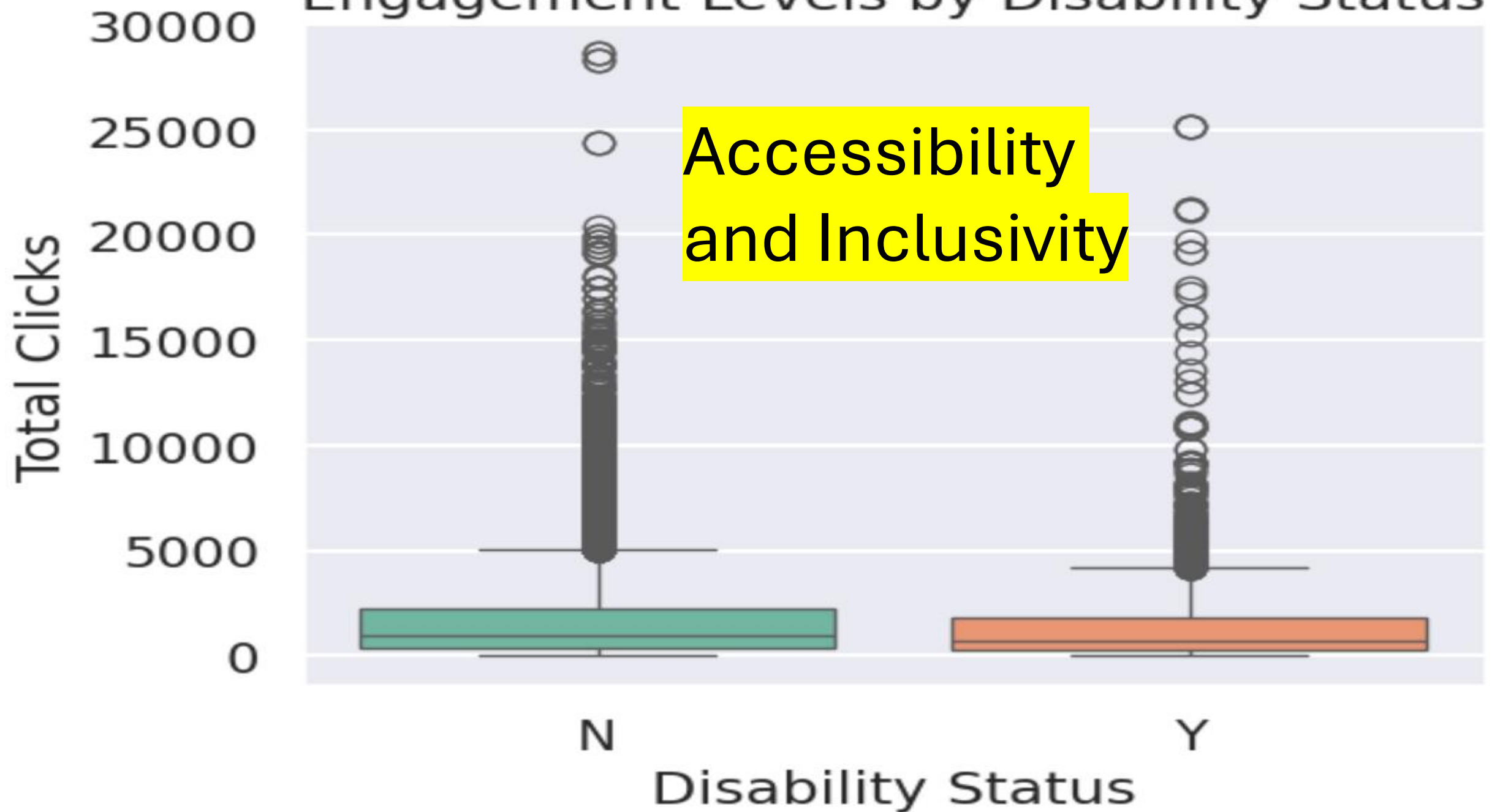
Using Exploratory Data Analysis, our goal is to:

- Identify the course outcomes, including the withdrawal and failure rates.
- Determine if the course is accessible and inclusive, regardless of student background or disability.
- Identify study habits that lead to course success.
- Identify efficient and inefficient course navigation that may impact student success

Course Grade Outcomes of Students



Engagement Levels by Disability Status



Engagement Levels by Disability Status

Total Clicks

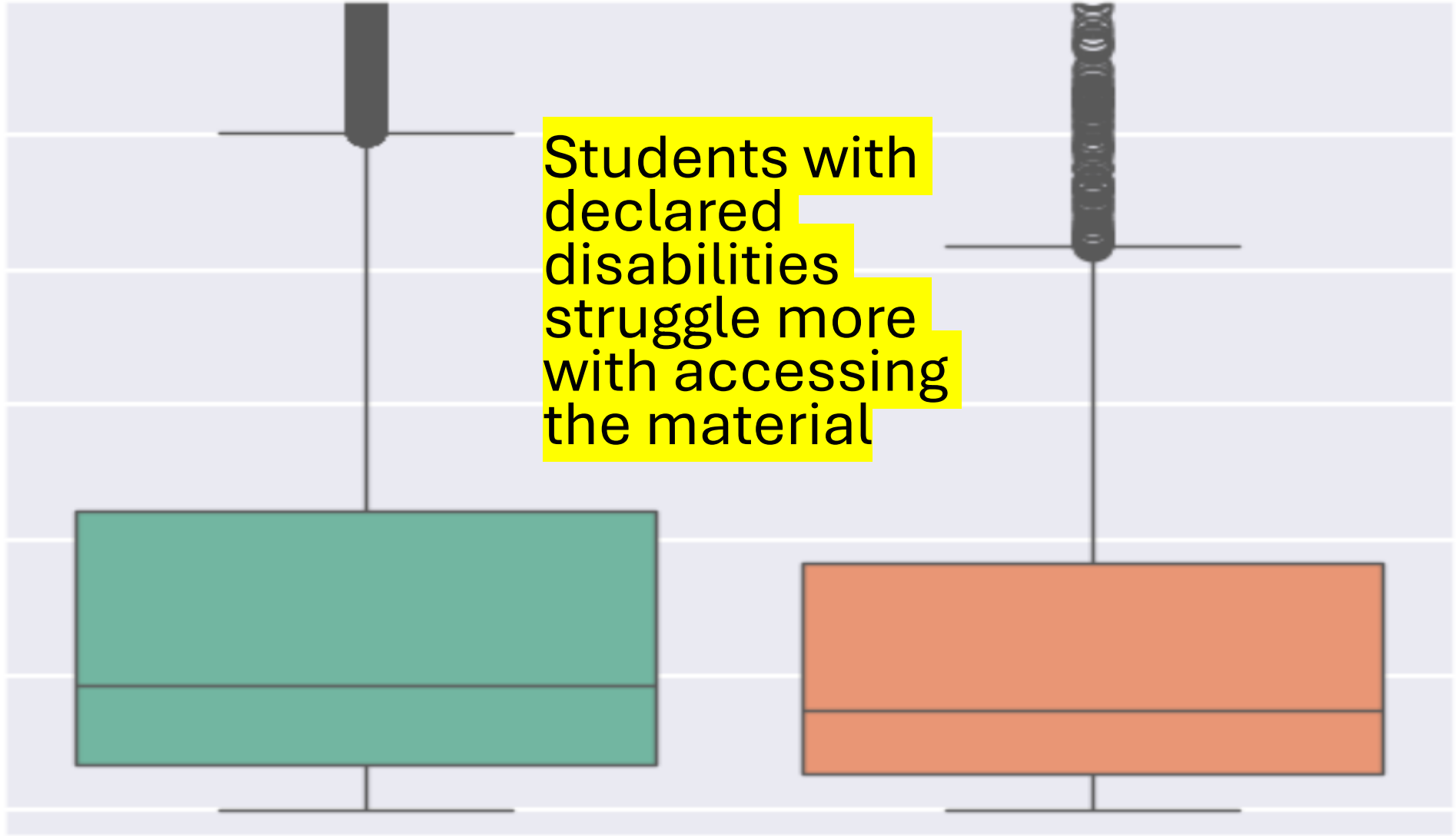
6000
5000
4000
3000
2000
1000
0

N

Y

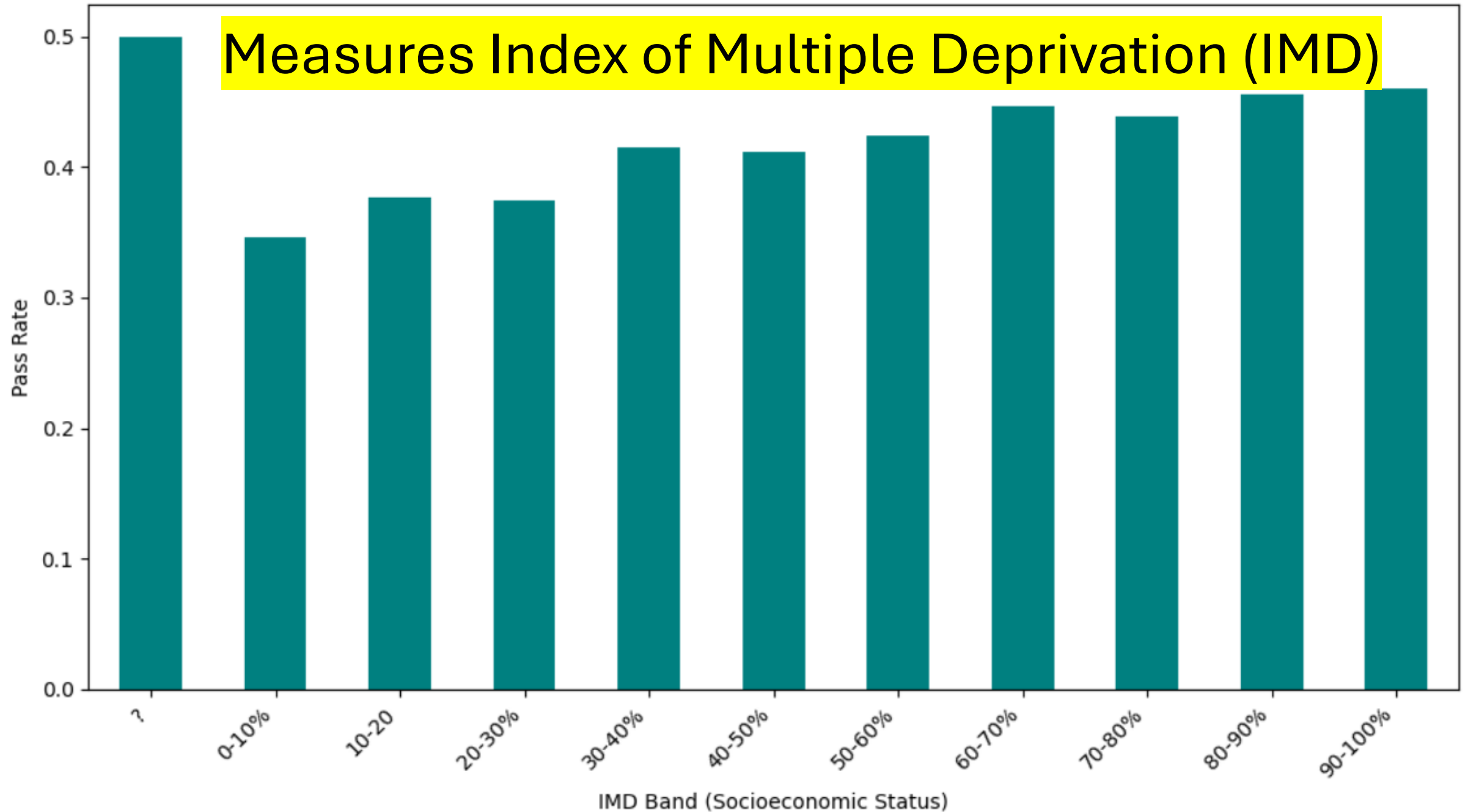
Disability Status

Students with
declared
disabilities
struggle more
with accessing
the material

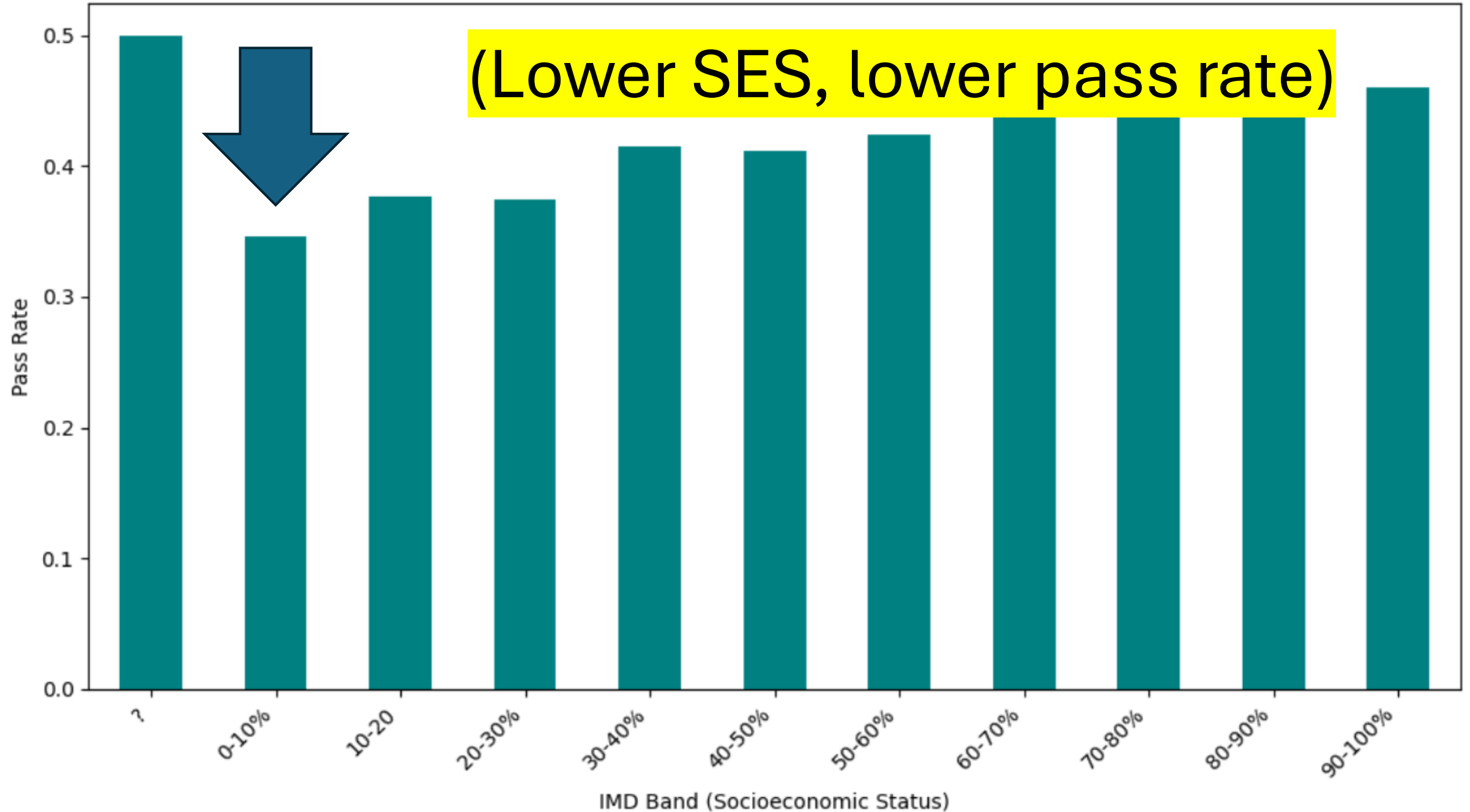


Pass Rate by Socioeconomic Status (IMD Band)

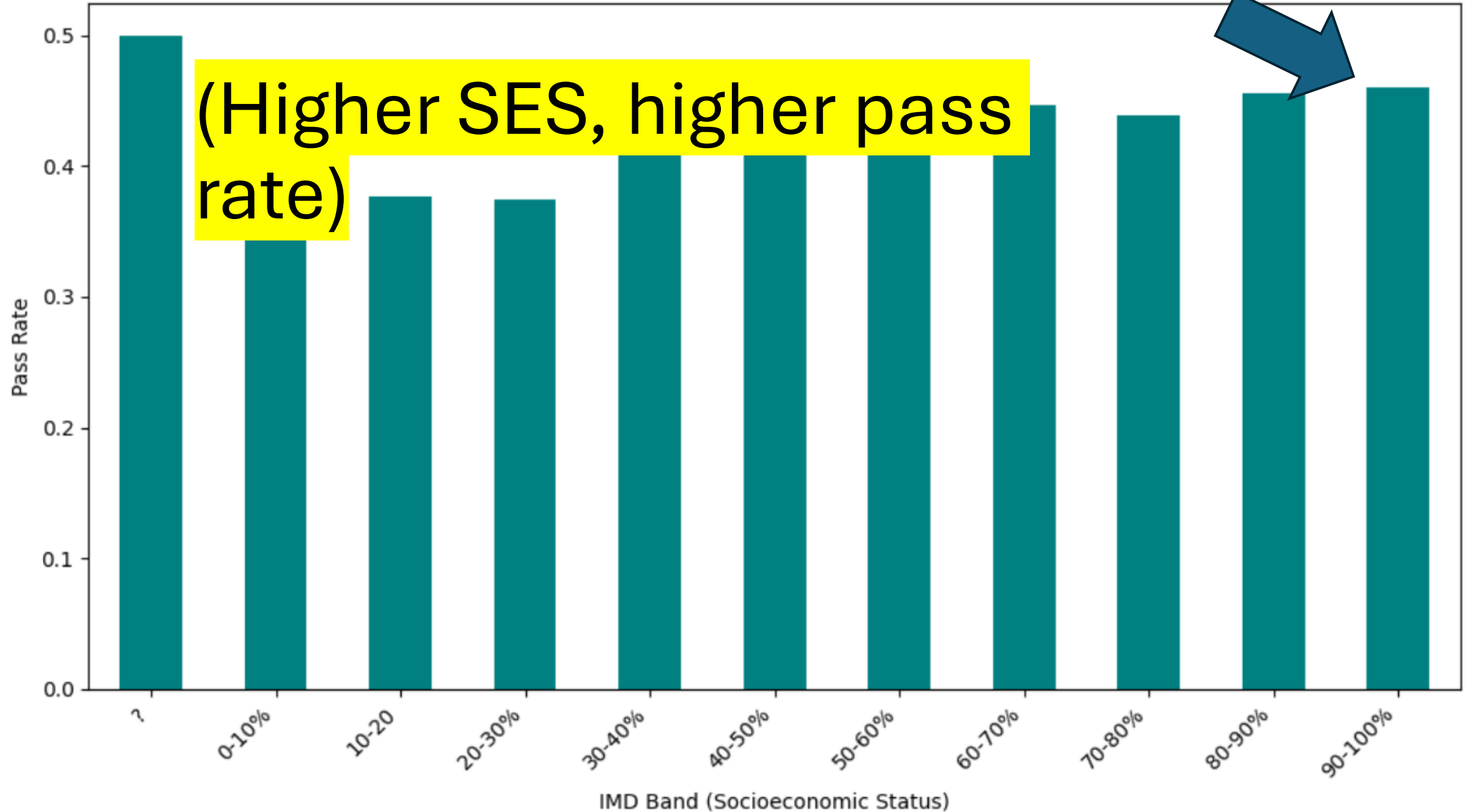
Measures Index of Multiple Deprivation (IMD)



Pass Rate by Socioeconomic Status (IMD Band)



Pass Rate by Socioeconomic Status (IMD Band)



Student Study Patterns by Day of Week

Day of Week

Mon

938,815

461,050

2,621,253

349,798

Tue

1,137,447

609,823

3,310,911

430,766

Wed

1,355,432

Successful students logged
in nearly twice as much as
those who struggled

Thu

1,335,642

445

Fri

1,245,359

540,704

Sat

1,123,947

663,808

3,328,700

419,322

Sun

927,637

450,664

2,550,597

333,384

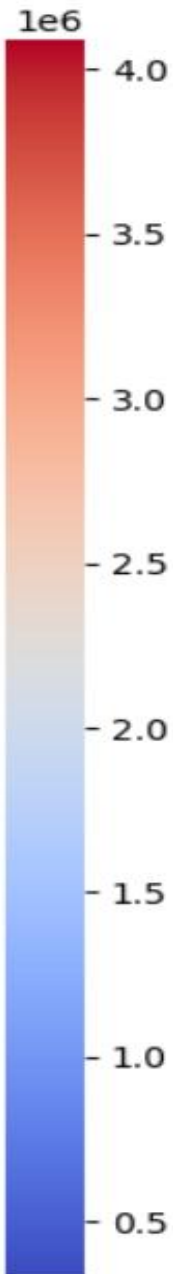
Distinction

Fail

Pass

Withdrawn

Total Logins per Day by Outcome



Student Study Patterns by Day of Week

Day of Week

Mon

938,815

461,050

2,621,253

349,798

Tue

1,137,447

609,823

3,310,911

430,766

Wed

1,355,432

Most successful students
carry their work into the
weekends

Thu

1,335,642

1,445

Fri

1,245,359

3,809,049

540,704

Sat

1,123,947

663,808

3,328,700

419,322

Sun

927,637

450,664

2,550,597

333,384

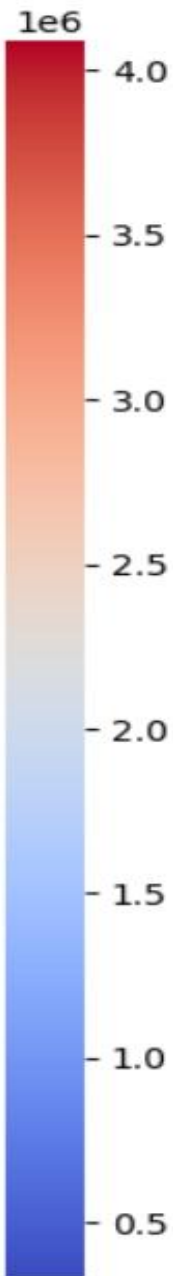
Distinction

Fail

Pass

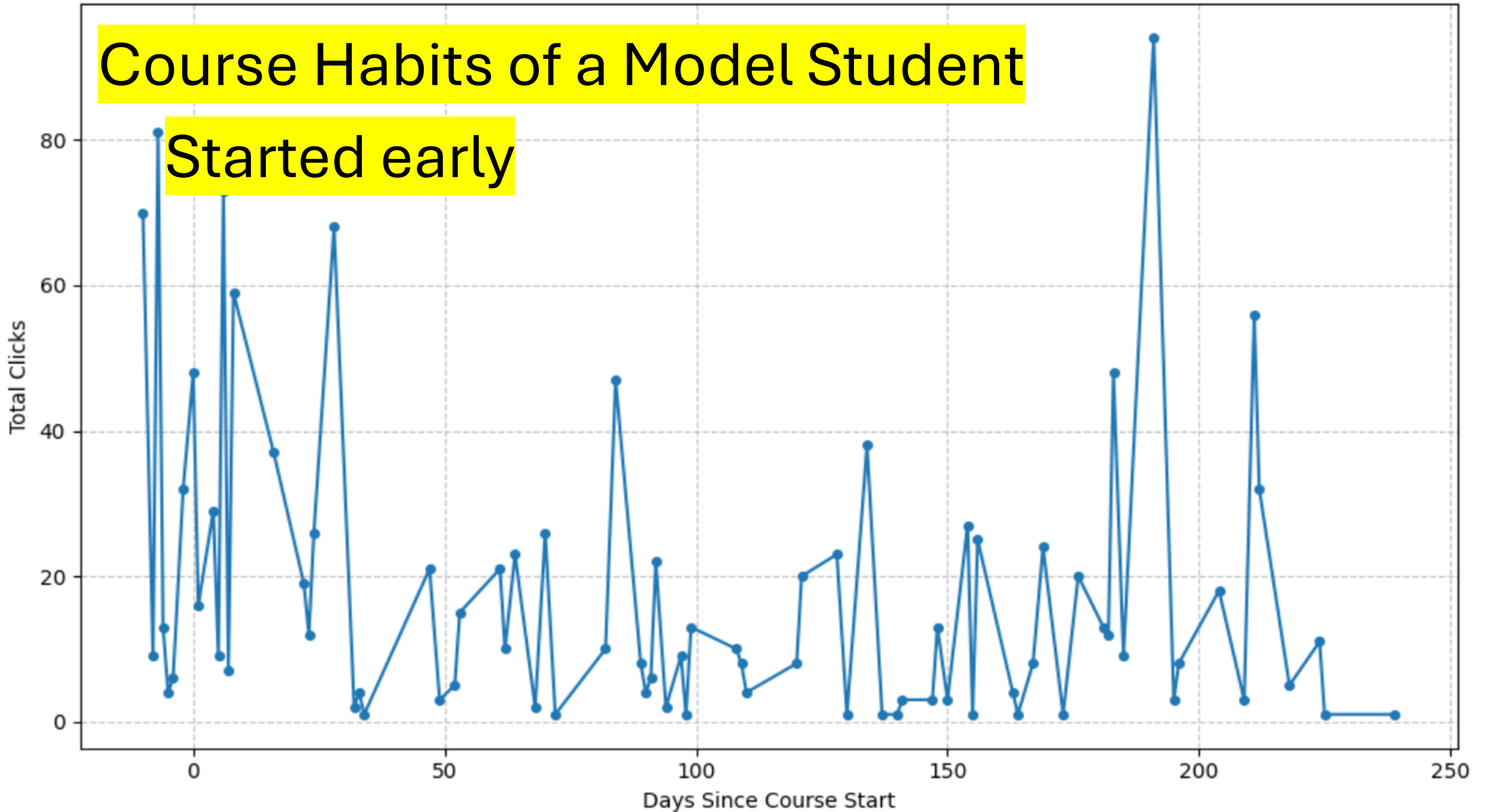
Withdrawn

Total Logins per Day by Outcome



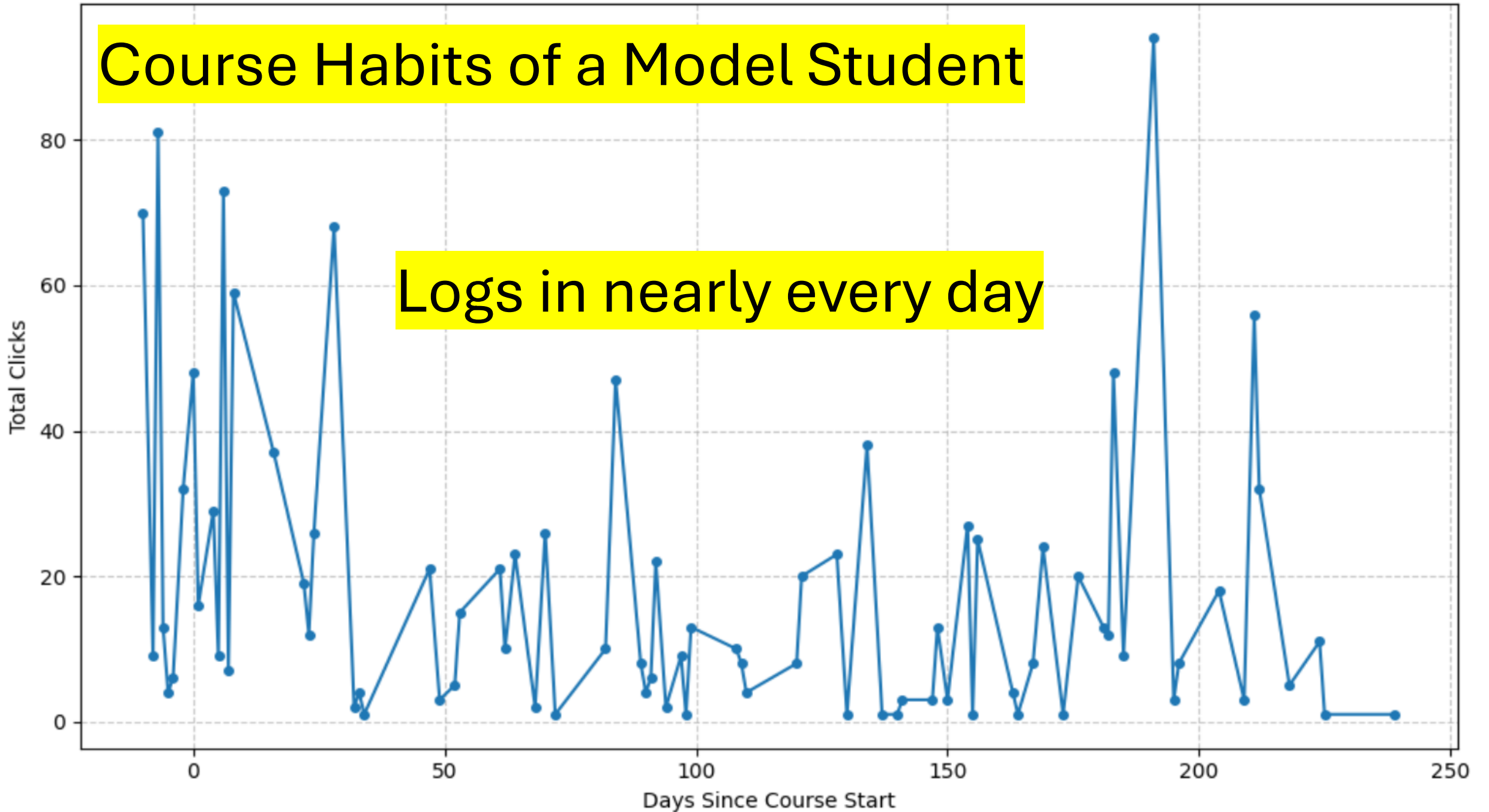
Course Habits of a Model Student

Started early

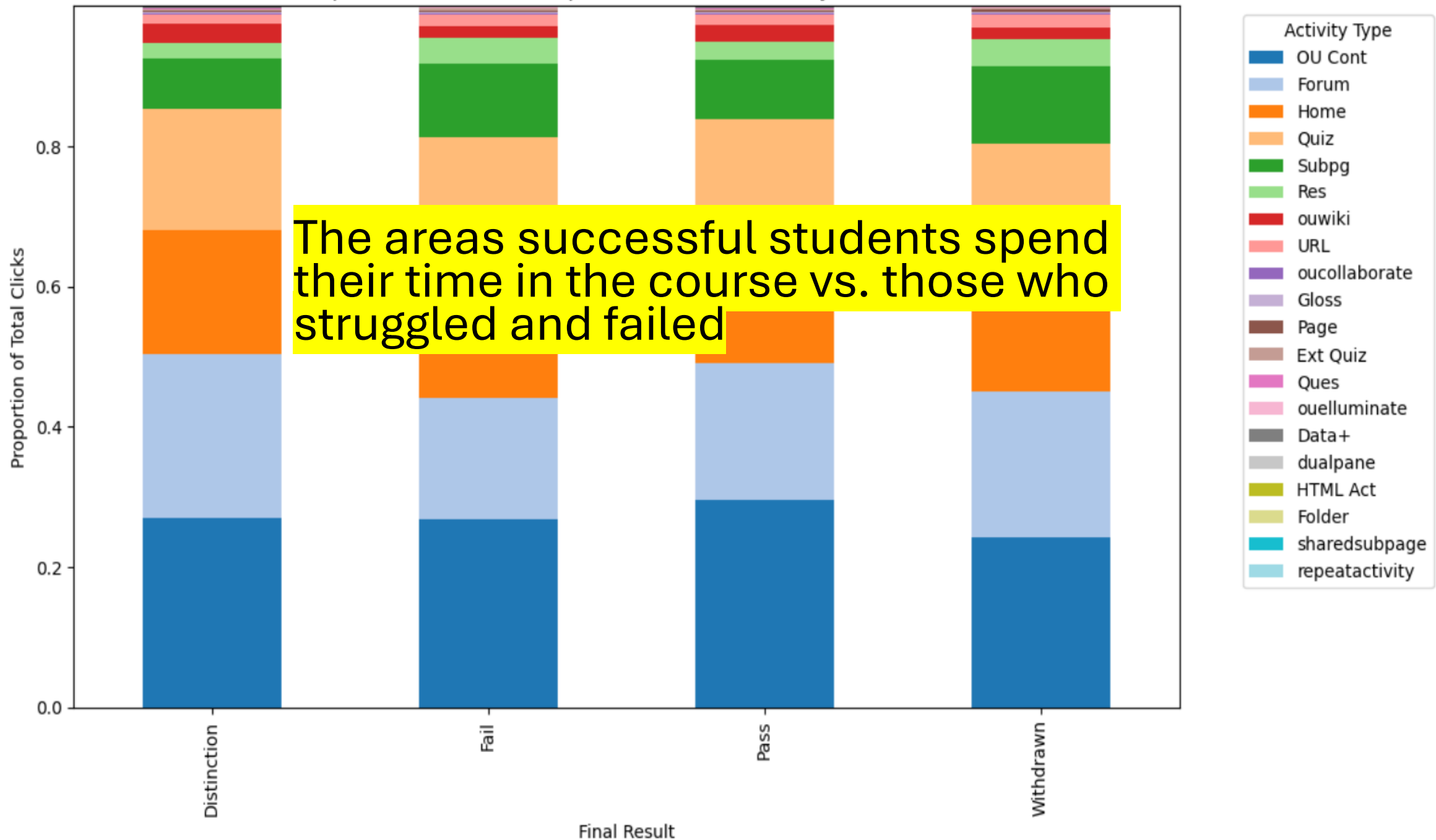


Course Habits of a Model Student

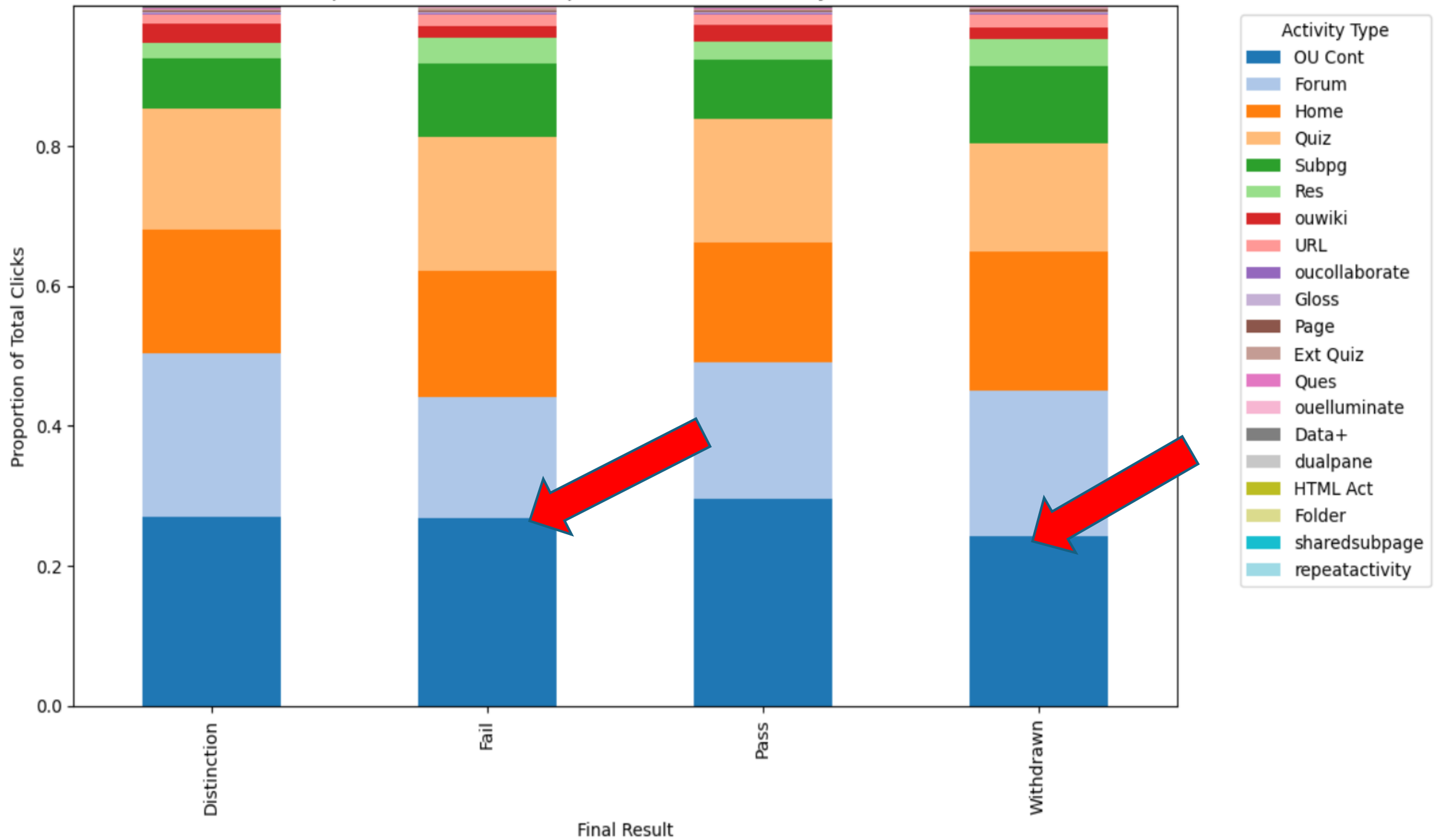
Logs in nearly every day



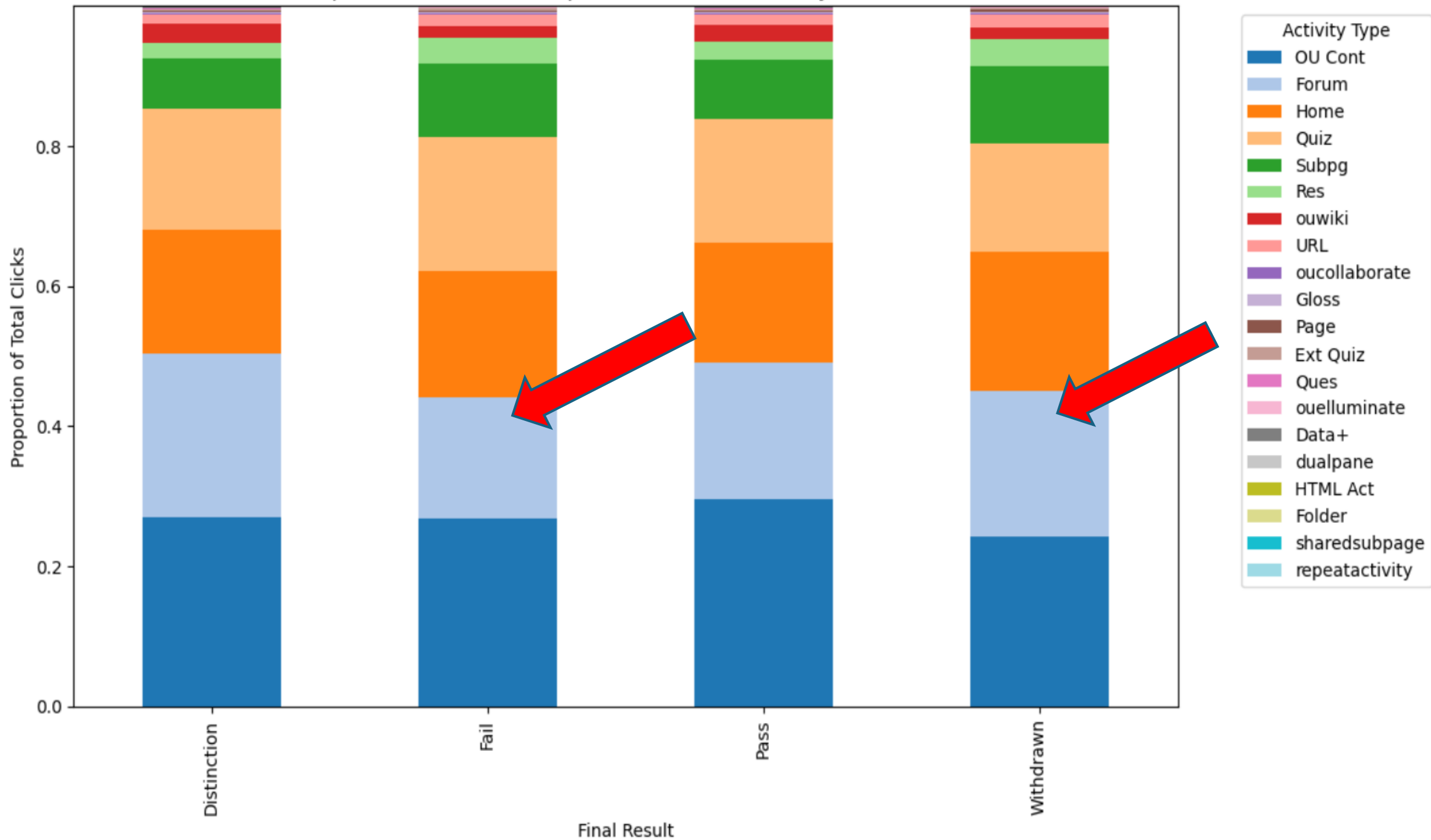
The areas successful students spend their time in the course vs. those who struggled and failed



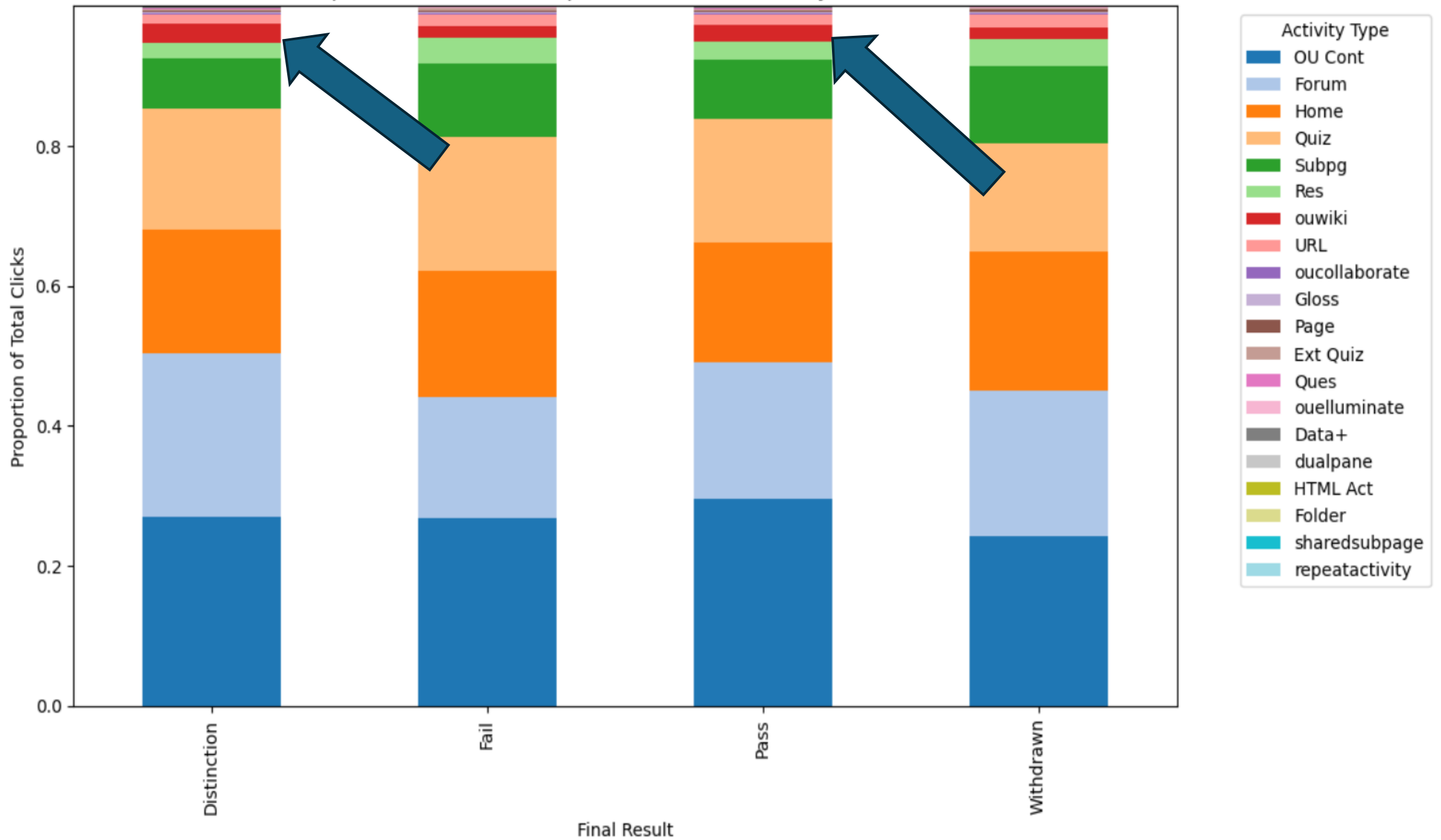
Proportion of Total Clicks Spent on Each Resource by Final Result



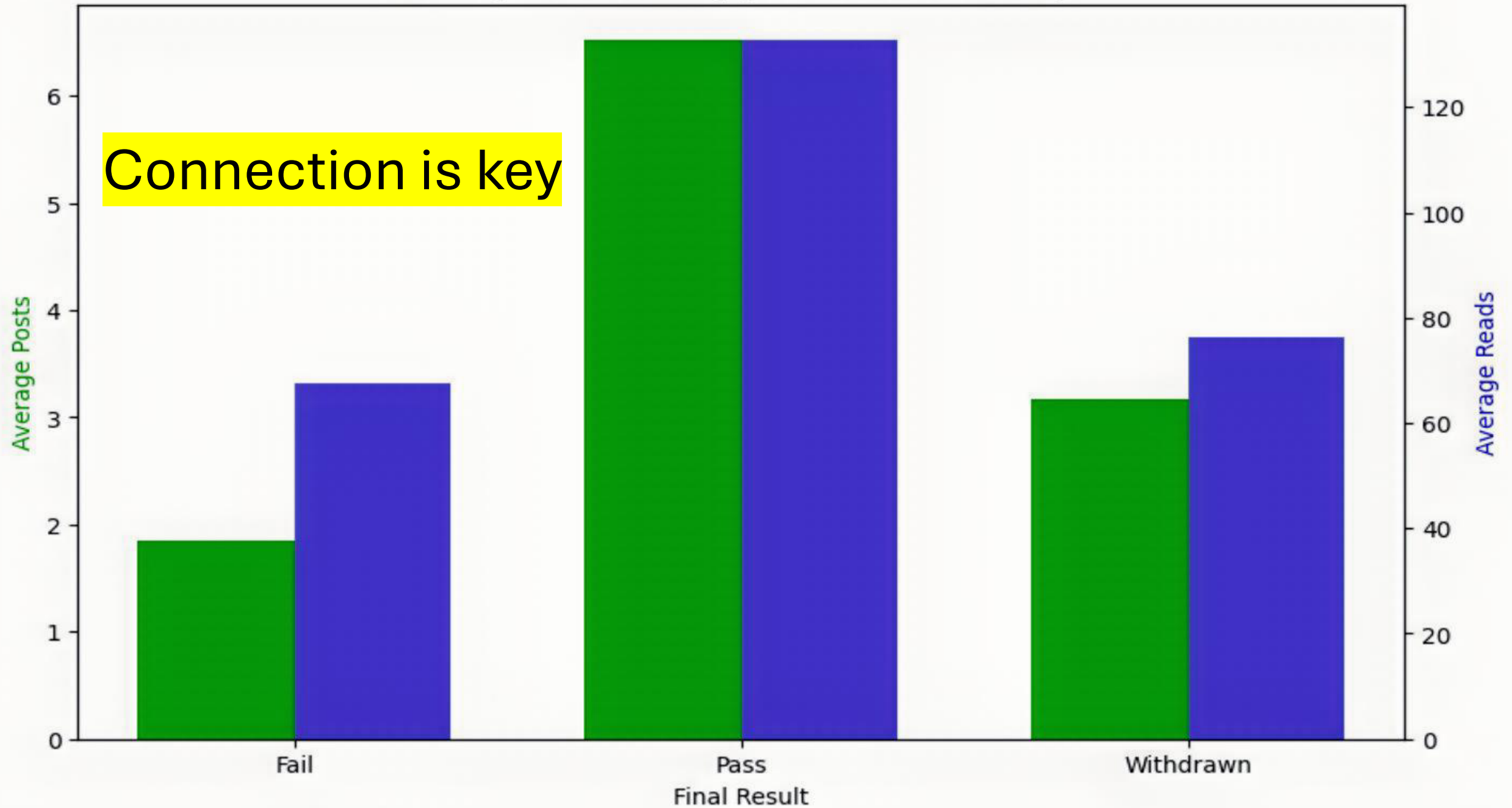
Proportion of Total Clicks Spent on Each Resource by Final Result



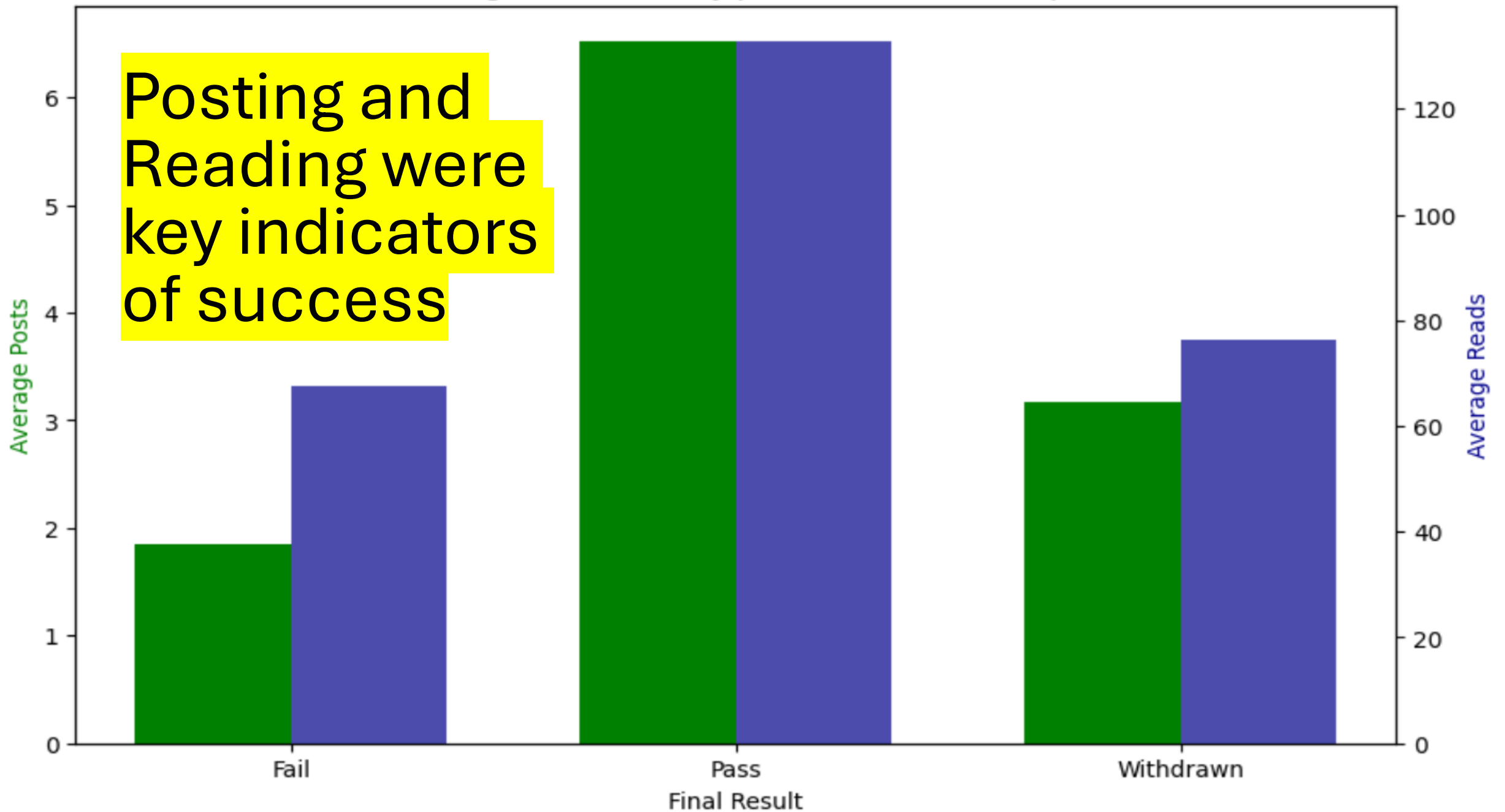
Proportion of Total Clicks Spent on Each Resource by Final Result



Average Social Activity per Performance Group

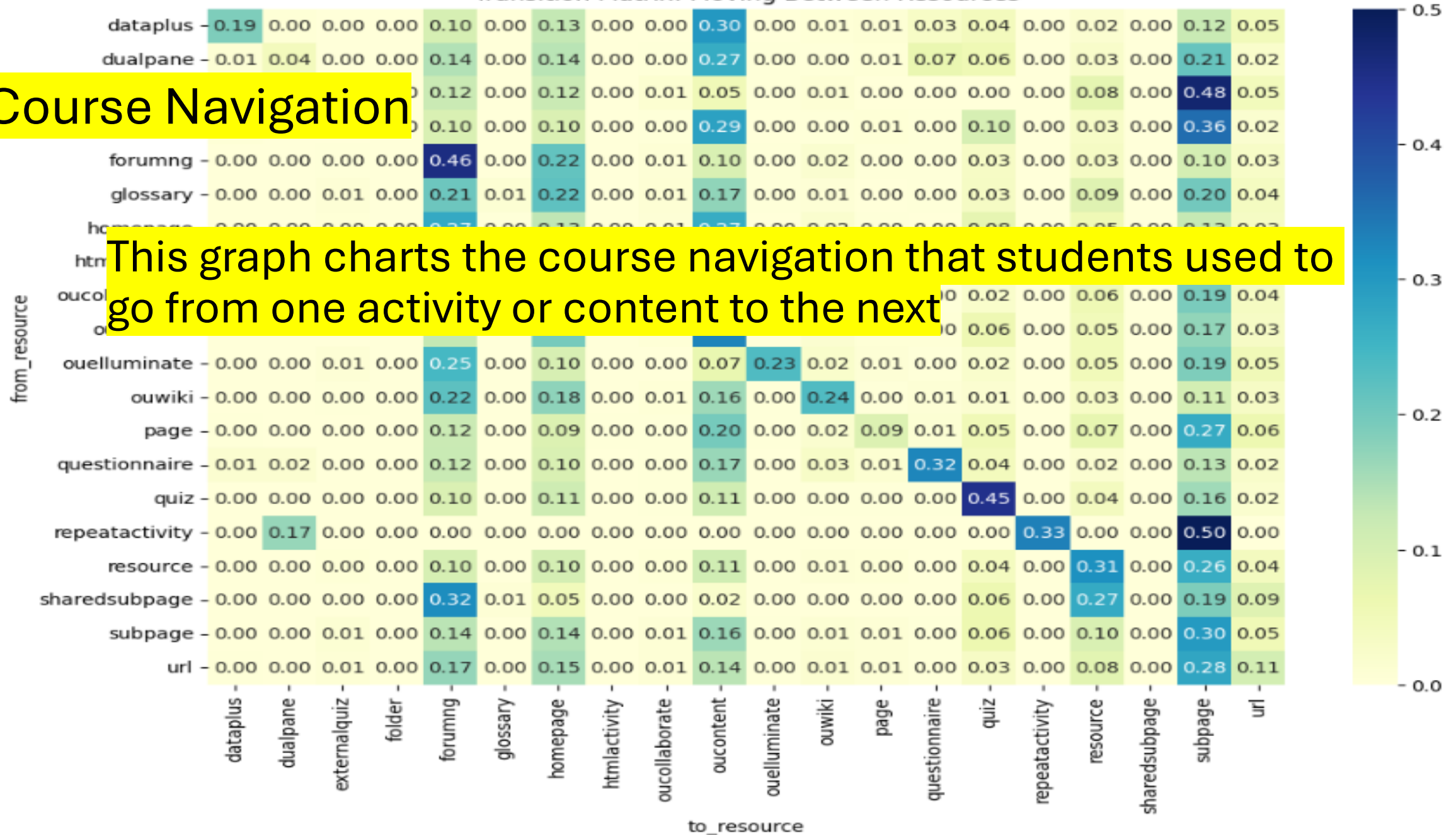


Average Social Activity per Performance Group



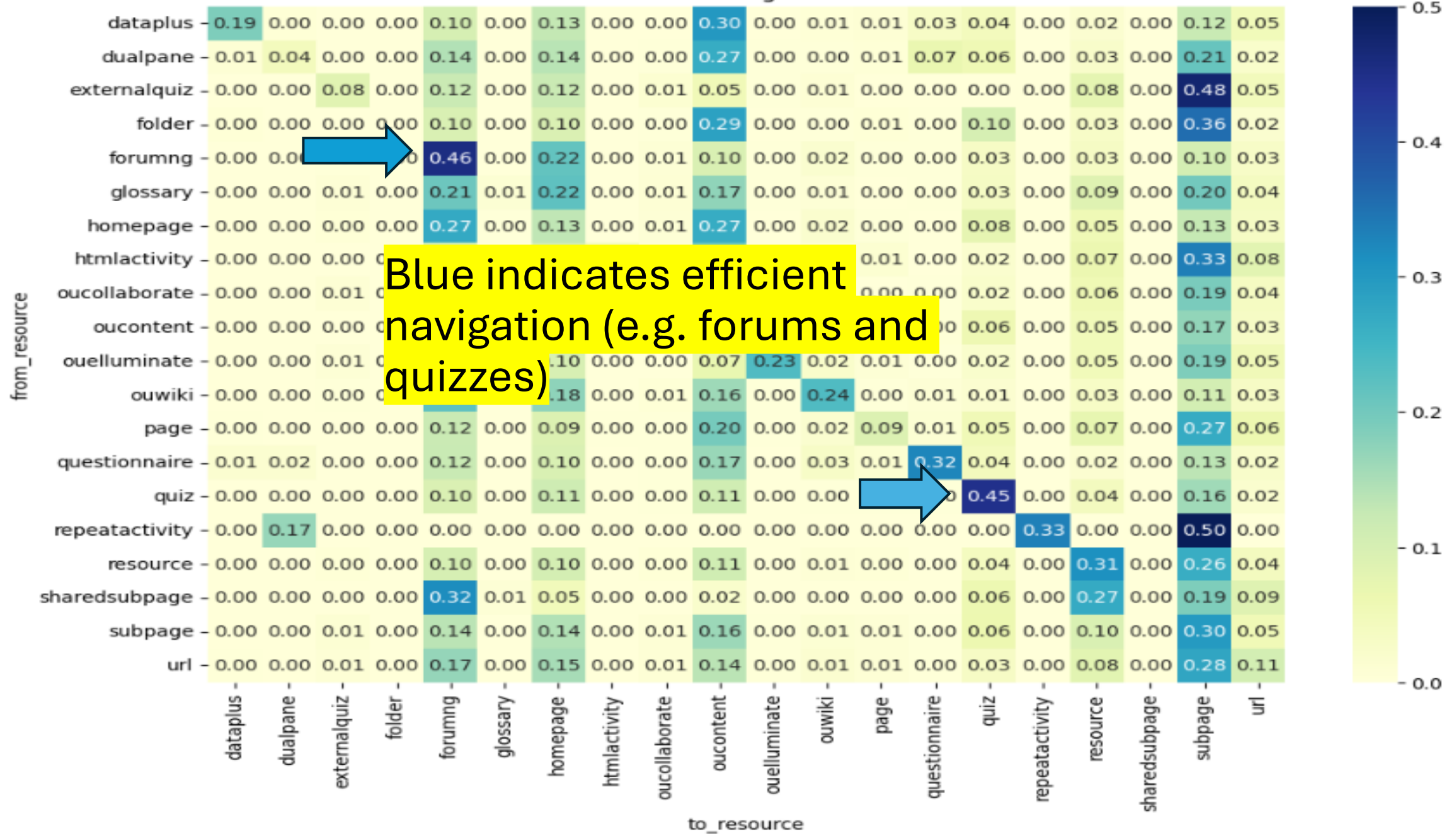
Course Navigation

Transition Matrix: Moving Between Resources

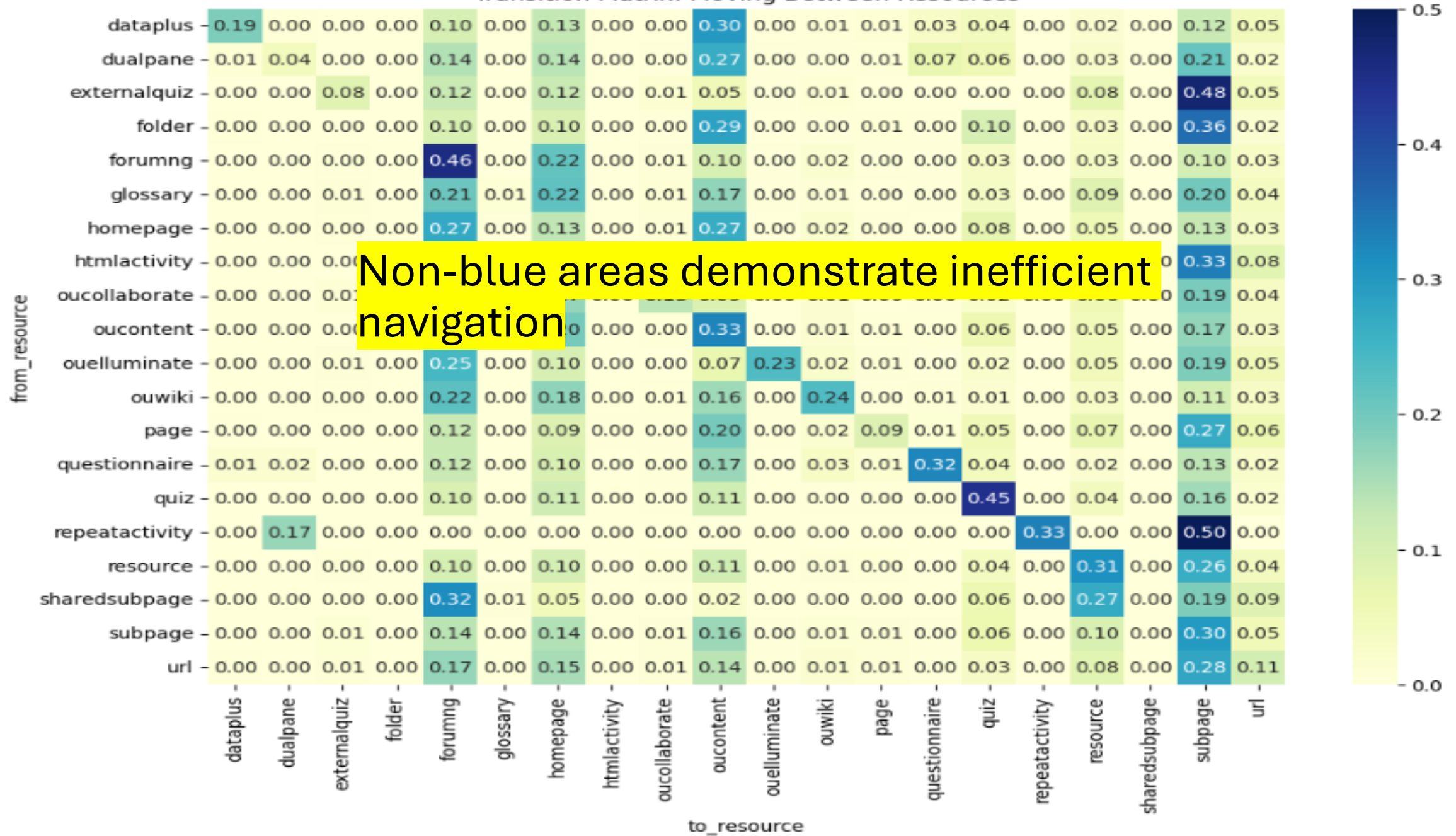


This graph charts the course navigation that students used to go from one activity or content to the next

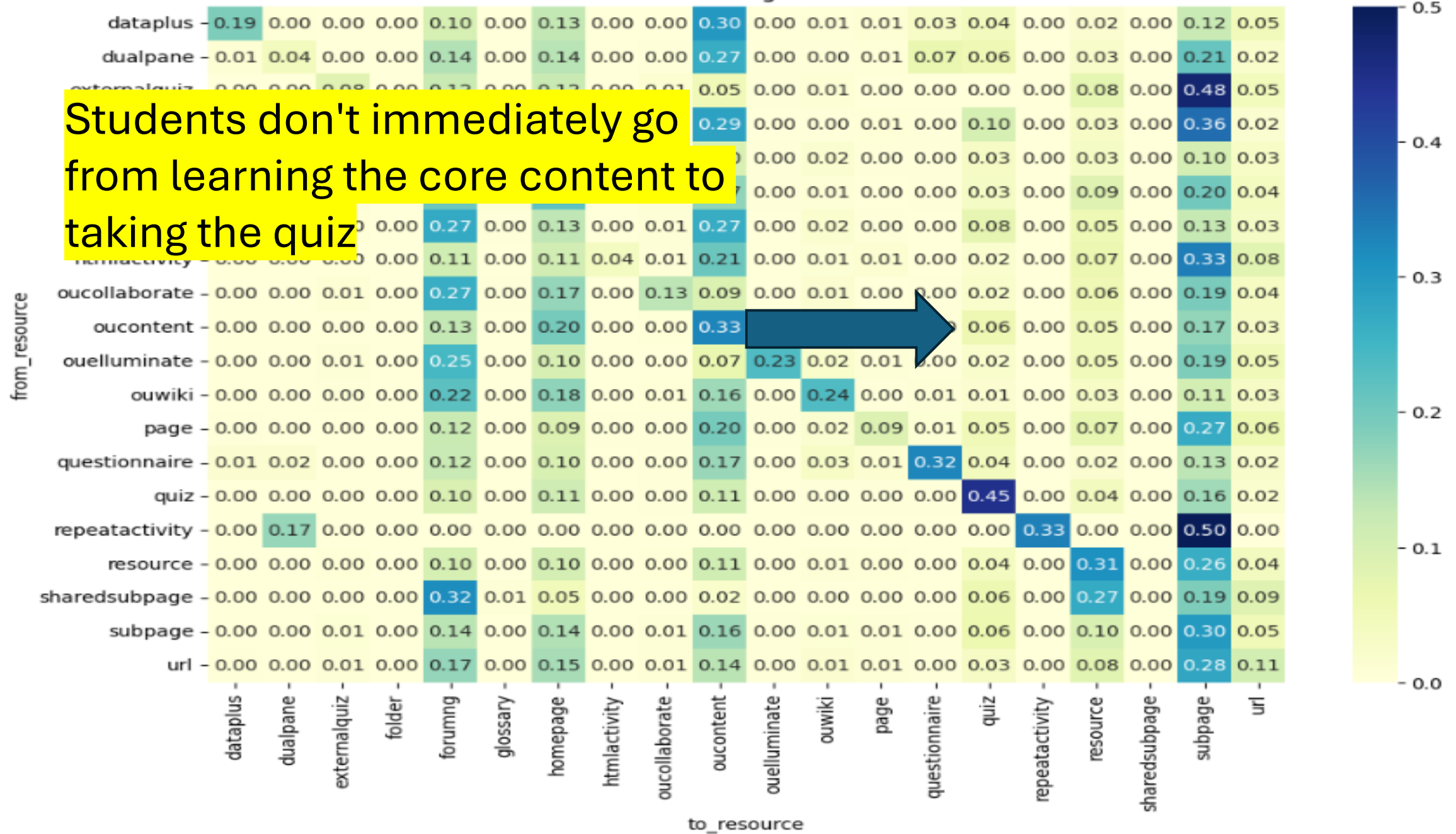
Transition Matrix: Moving Between Resources



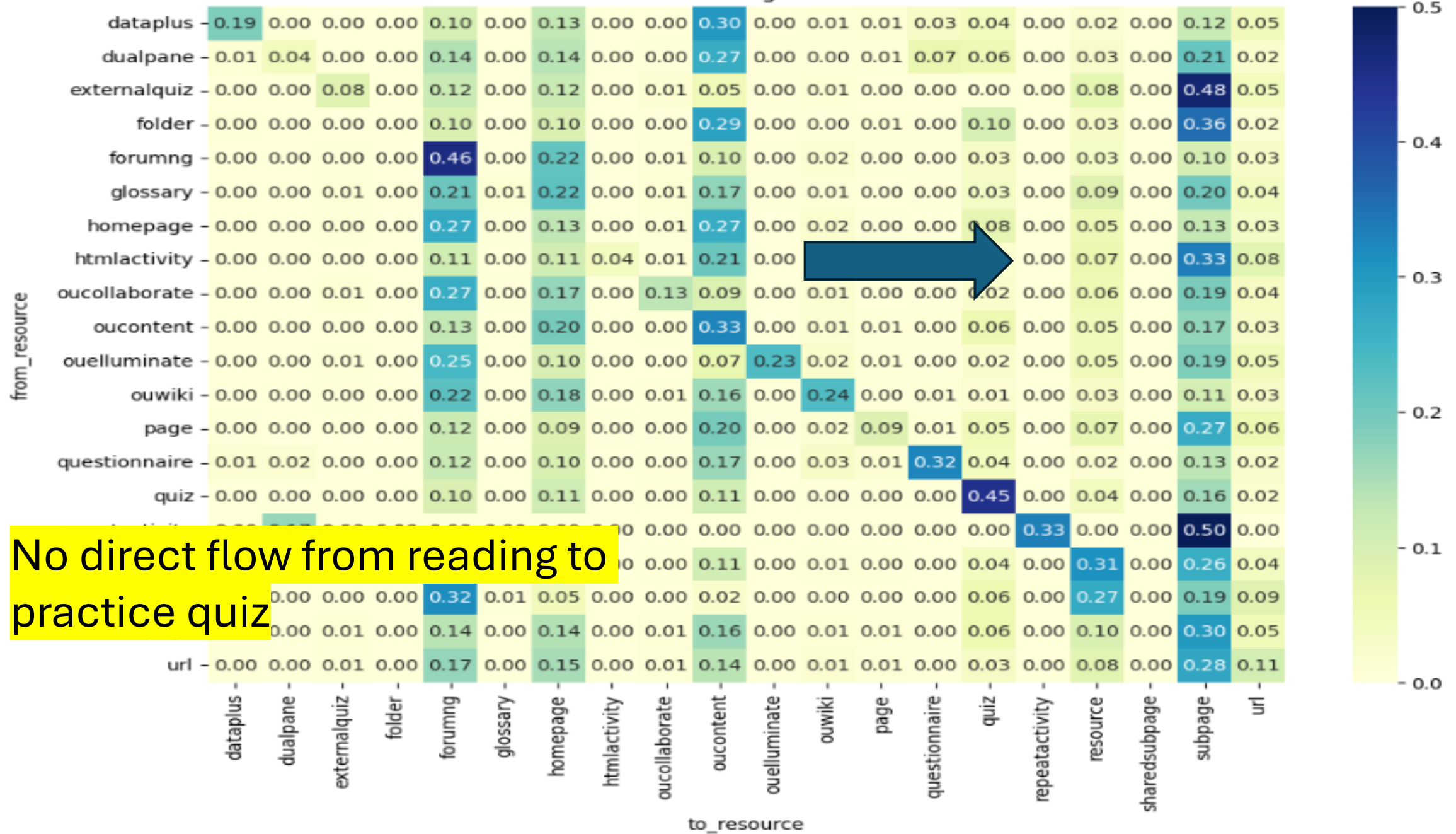
Transition Matrix: Moving Between Resources



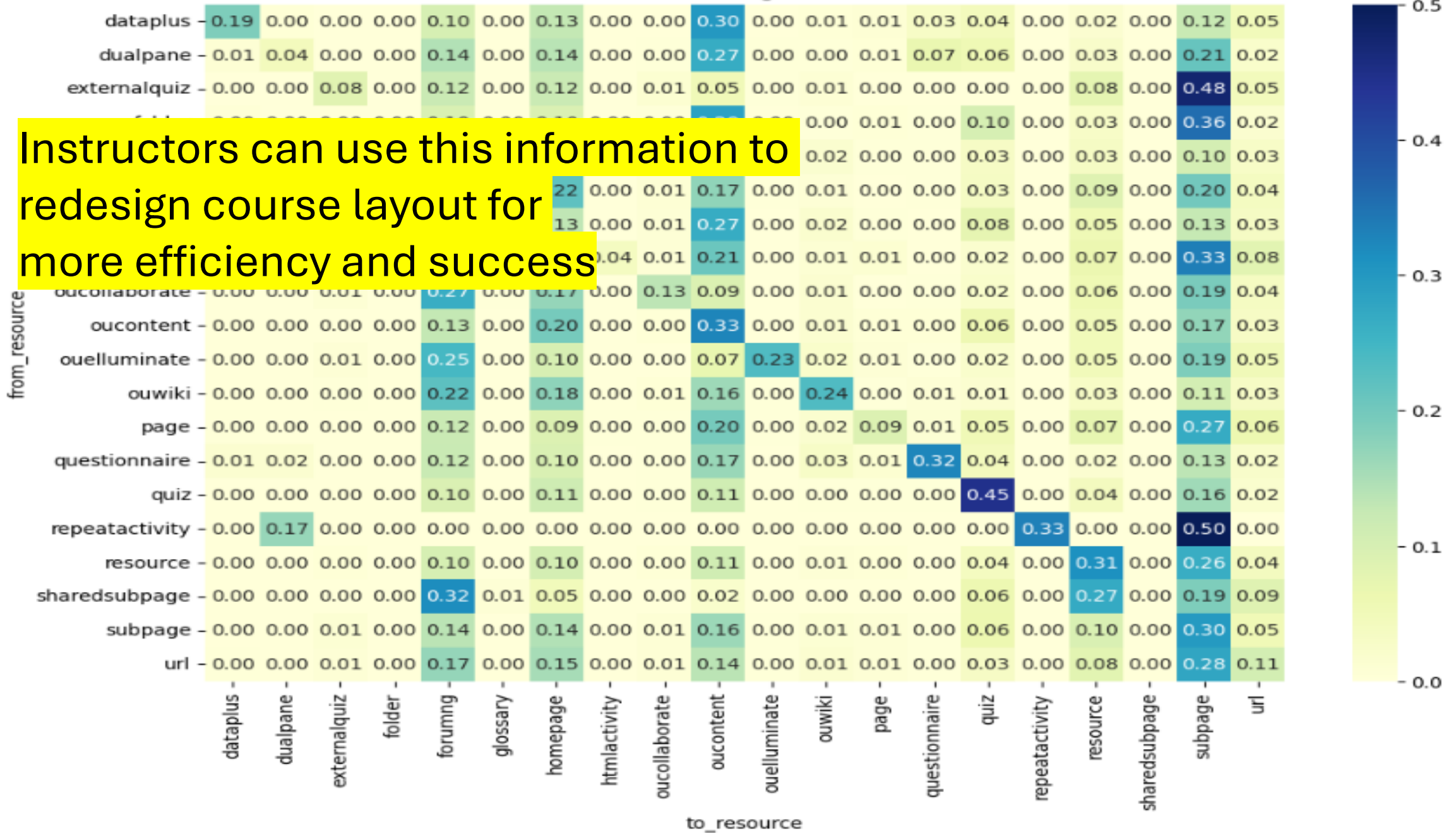
Transition Matrix: Moving Between Resources



Transition Matrix: Moving Between Resources



Transition Matrix: Moving Between Resources



From Analysis to Prediction

- Use early-course data from the first 20% of the course
- Predict student outcomes before the course ends
- Enable early identification of at-risk students

Machine Learning Approach

- Learn patterns from past students
- Apply patterns to new students
- Compare multiple models
- Select best-performing method

Prediction Tasks

- Multiclass classification
 - Distinction
 - Pass
 - Fail
 - Withdrawn
- Binary classification
 - At-risk
 - Not at-risk

Features Used for Prediction

- Early engagement
 - Total clicks
 - Active days
- Early assessment performance
 - Average score
 - Max score
- Student information
 - Education
 - Age
 - Region
- Registration behavior

Models Tested

-Logistic Regression

-Random Forest

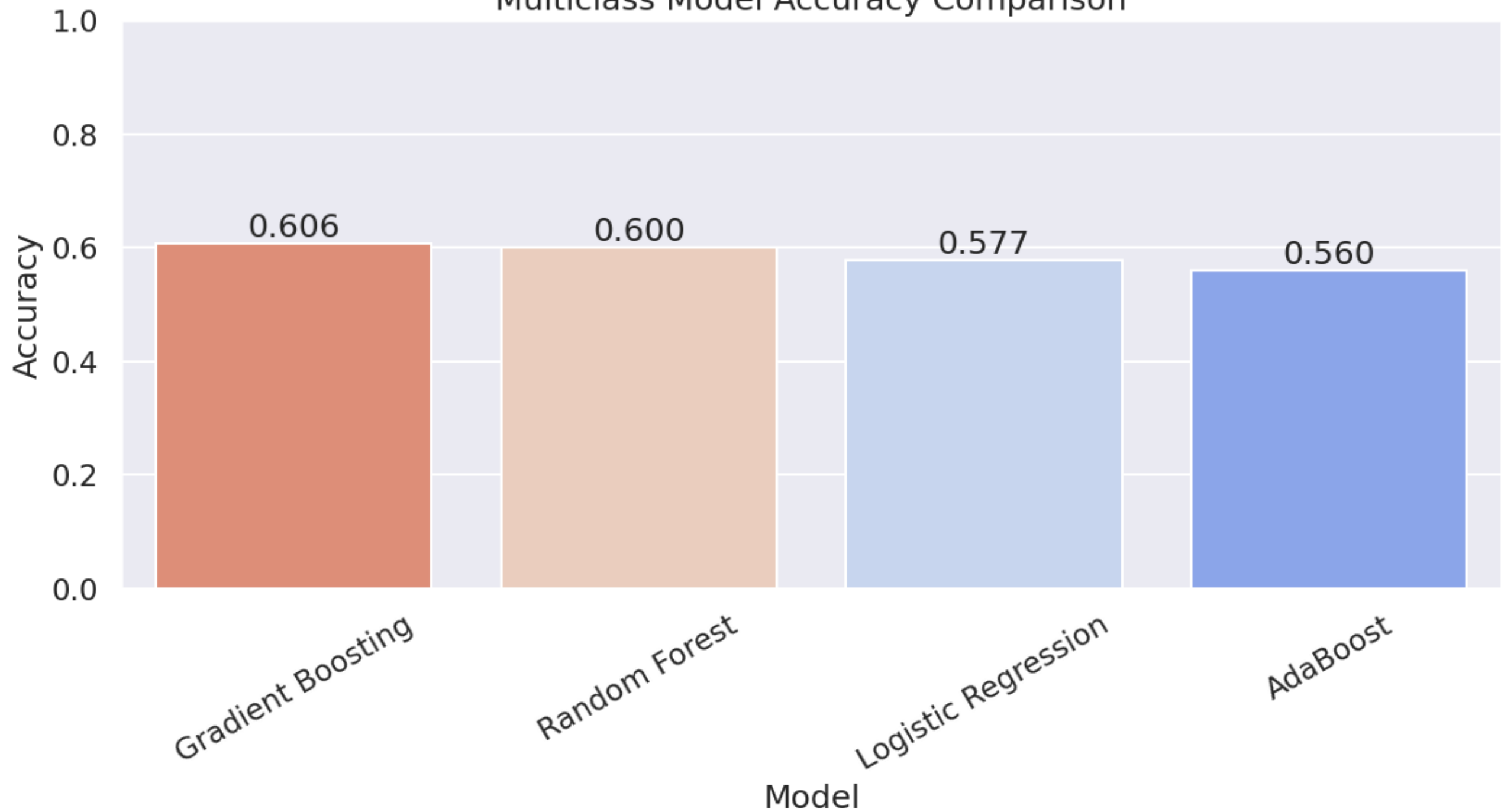
-Gradient Boosting

-AdaBoost

Multiclass Prediction Results

- Compare multiple machine learning models
- Evaluate performance
- Show highest accuracy model

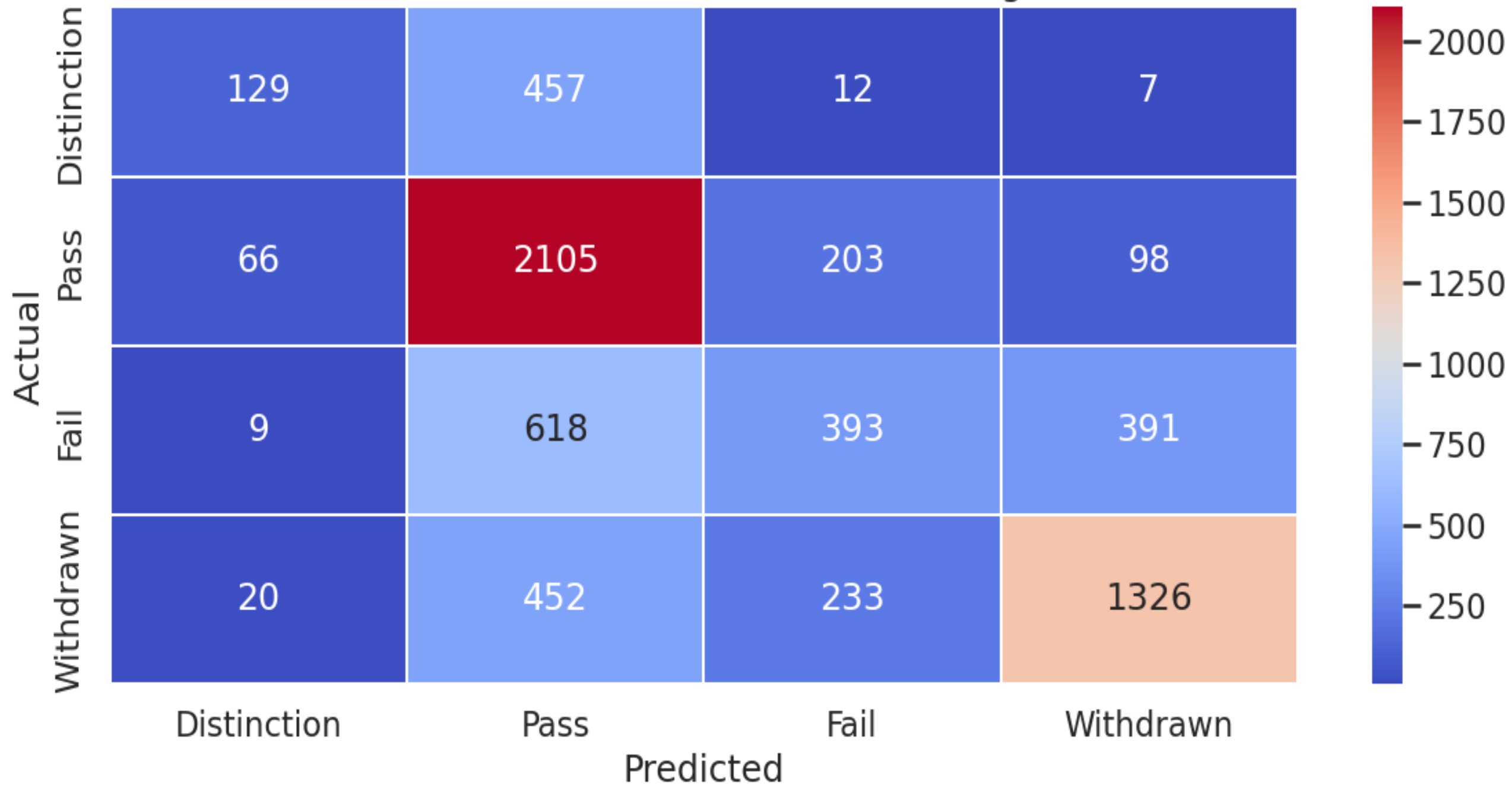
Multiclass Model Accuracy Comparison



Multiclass Confusion Matrix

- Breaks predictions down by class
- Shows correct vs. incorrect classifications
- Helps visualize where the model is making mistake

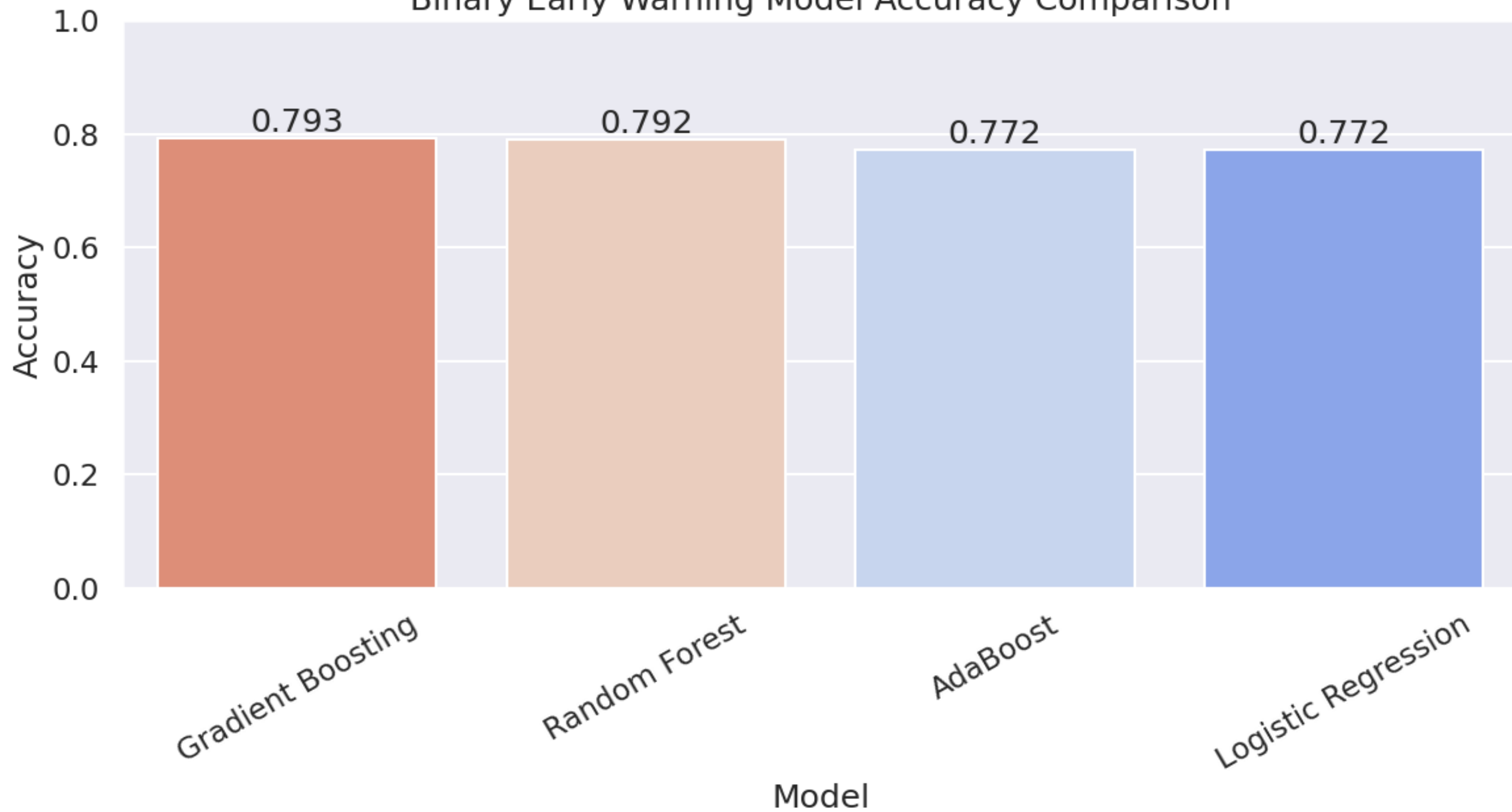
Confusion Matrix - Gradient Boosting



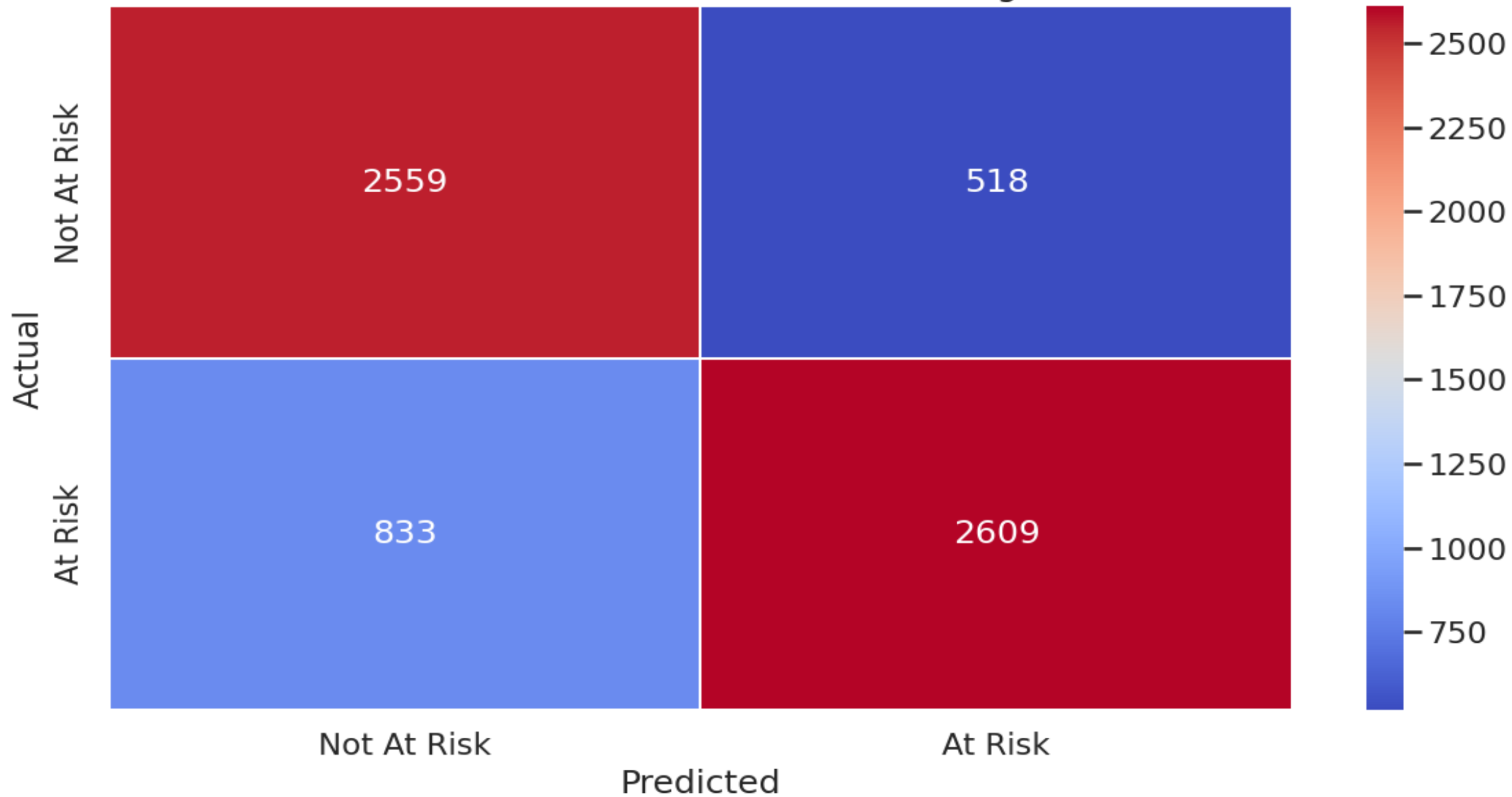
Binary Classification Results

- Binary framing removes unnecessary categories
- Highlights students who may fail or withdraw
- Makes the model's decisions simpler to interpret and act on

Binary Early Warning Model Accuracy Comparison



Confusion Matrix - Gradient Boosting

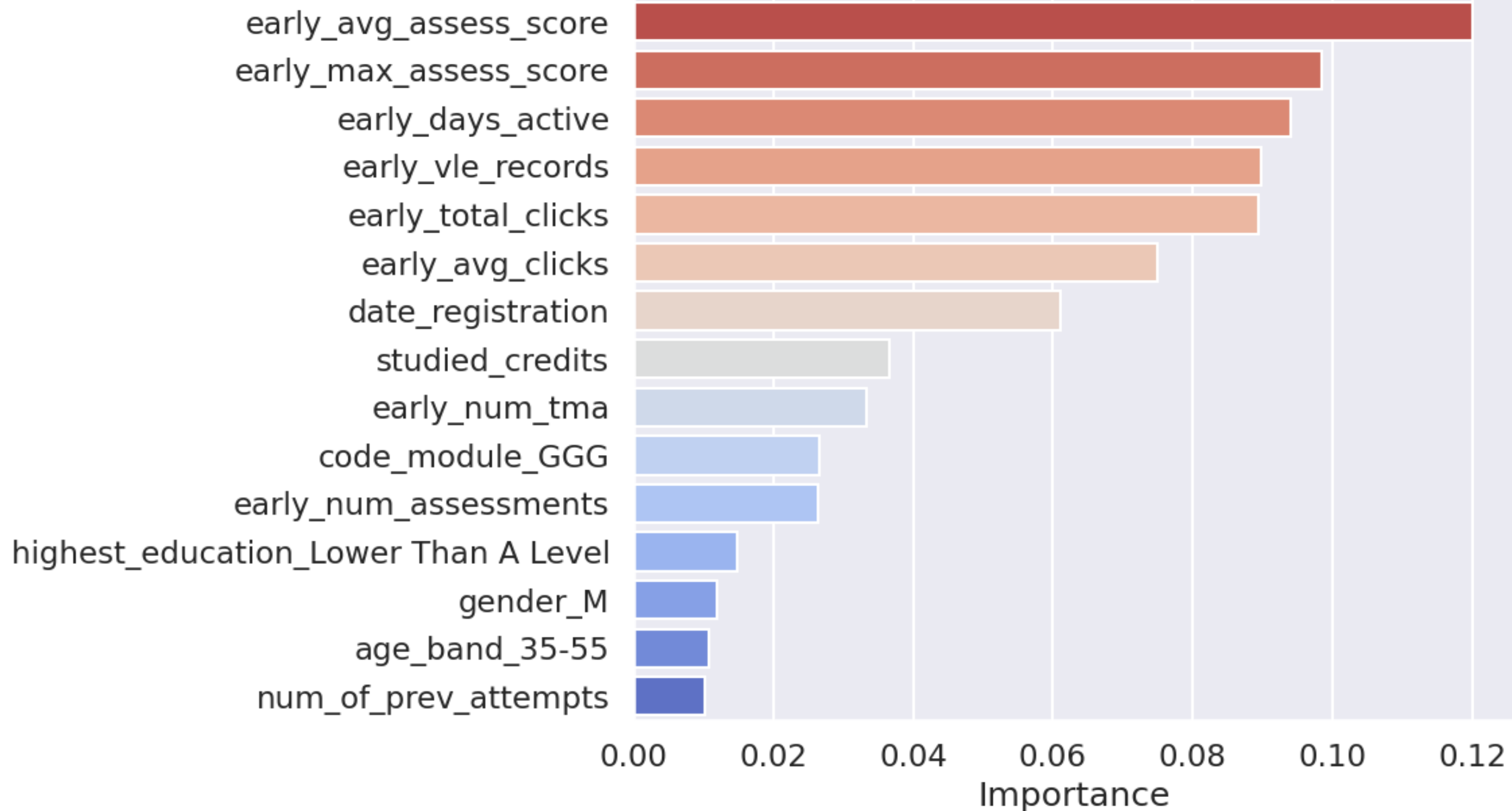


Feature Importance

- What drives predictions
- Key features
- Explain model results
- Insights from variables

Top 15 Important Features for Early Risk Prediction

Feature



Causal Effects: From Correlation to Causation

Correlation: Students who click more tend to succeed

Limitation: Correlation does not imply causation

Key Question:

- Does engagement *cause* success?
- What happens if we increase engagement?
- Approach:
 - Compare similar students
 - Control for background factors

Causal Effect of Engagement on Success

Treatment: High engagement (high clicks)

Outcome: Passing the course

Result (Average Treatment Effect (ATE)):

- +32.3% increase in pass probability

Interpretation:

- Increasing engagement improves success

Implication:

- Early intervention strategies can be effective

Final Machine Learning Insight

- **Early behavior predicts later outcomes**
- **Engagement has a causal effect on success**
- **Engagement and assessment matter most**
- **Binary prediction is practical and useful**
- **Machine learning can support intervention**

Sources

Kuzilek, J., Hlosta, M. & Zdrahal, Z. (2017) Open University Learning Analytics dataset. *Sci Data*, 4, 170171

<https://doi.org/10.1038/sdata.2017.171>

The Open University. (n.d.). *Facts and figures*.

<https://www.open.ac.uk/policies-and-reports/facts-and-figures>
(Accessed April 13, 2026)

Sim, S. (2024). *Analysis of the OULAD dataset*. GitHub Pages.

<https://sergiosim.github.io/OULAD/>

Thank you!

Questions?
